

Security Scan Report

Target: C:\Users\Dhruv\Desktop\College Stuff\Projects\Messenger app

Scan Date: July 23, 2026 at 13:49 UTC

Scan ID: e961216c-21ff-4d46-8ca5-4f548f6e2b7e

0 CRITICAL

0 HIGH

2 MEDIUM

1 LOW

Executive Summary

Security scan of 'C:\Users\Dhruv\Desktop\College Stuff\Projects\Messenger app' identified 6 raw findings, reduced to 3 unique issues after deduplication. 2 MEDIUM-priority issue(s) should be addressed in the current development sprint. 1 LOW-priority issue(s) have been logged for backlog tracking.

Scan Statistics

Metric	Value
Total Raw Findings	6
After Deduplication	3
Critical Findings	0
High Findings	0
Medium Findings	2
Low Findings	1
Scan Duration	14.0s

Detailed Findings

[MEDIUM] #1: Hardcoded Secret in C:\Users\Dhruv\Desktop\College Stuff\Projects\Messenger app\backend\nexus-data.json:7

Risk Score	0.5488
Affected Location	C:\Users\Dhruv\Desktop\College Stuff\Projects\Messenger app\backend\nexus-data.json:7
Vulnerability Type	HARDCODED_SECRET
CWE ID	CWE-798: Use of Hard-coded Credentials
Detected By	semgrep

Explanation
Hardcoded secrets (passwords, API keys, tokens) in source code are a critical security risk. Anyone with access to the codebase can extract and misuse these credentials.

Code or HTTP Excerpt

```
>>> 5 |      "username": "alice",
      6 |      "email": "alice@test.com",
      7 |      "password_hash": "$2a$12$HHgdJ2zQQA/L98bwDj5F2069e9UHktwVEysZc6CVLY.rqR.ENnkI.",
      8 |      "display_name": "Alice",
      9 |      "avatar_url": "",
```

Business Impact
Exposed credentials could allow unauthorized access to external services, databases, or cloud infrastructure.

Fix Recommendation
Rule-based recommendation: Move all secrets to environment variables. Use python-decouple or python-dotenv to load from .env files. Add .env to .gitignore. Rotate any credentials that were previously committed.

LLM recommendation: Move all secrets to environment variables. Use python-decouple or python-dotenv to load from .env files. Add .env to .gitignore. Rotate any credentials that were previously committed.

[MEDIUM] #2: Hardcoded Secret in C:\Users\Dhruv\Desktop\College Stuff\Projects\Messenger app\frontend\release\win-unpacked\resources\app\backend\nexus-data.json:7

Risk Score	0.5488
Affected Location	C:\Users\Dhruv\Desktop\College Stuff\Projects\Messenger app\frontend\release\win-unpacked\resources\app\backend\nexus-data.json:7
Vulnerability Type	HARDCODED_SECRET
CWE ID	CWE-798: Use of Hard-coded Credentials
Detected By	semgrep

Explanation
Hardcoded secrets (passwords, API keys, tokens) in source code are a critical security risk. Anyone with access to the codebase can extract and misuse these credentials.

Code or HTTP Excerpt

```
5 |      "username": "alice",
6 |      "email": "alice@test.com",
>>> 7 |      "password_hash": "$2a$12$HHgdJ2zQQA/L98bwDj5F2069e9UHktwVEysZc6CVLY.rqR.ENnkI.",
8 |      "display_name": "Alice",
9 |      "avatar_url": "",
```

Business Impact

Exposed credentials could allow unauthorized access to external services, databases, or cloud infrastructure.

Fix Recommendation

Rule-based recommendation: Move all secrets to environment variables. Use python-decouple or python-dotenv to load from .env files. Add .env to .gitignore. Rotate any credentials that were previously committed.

LLM recommendation: Move all secrets to environment variables. Use python-decouple or python-dotenv to load from .env files. Add .env to .gitignore. Rotate any credentials that were previously committed.

[LOW] #3: Unknown in C:\Users\Dhruv\Desktop\College Stuff\Projects\Messenger app\backend\fileRoutes.js:136

Risk Score	0.3384
Affected Location	C:\Users\Dhruv\Desktop\College Stuff\Projects\Messenger app\backend\fileRoutes.js:136
Vulnerability Type	UNKNOWN
CWE ID	CWE-22: Improper Limitation of a Pathname to a Restricted Directory ('Path Traversal')
Detected By	semgrep

Explanation

A security vulnerability was detected in your code. This issue could potentially be exploited by attackers to compromise the security of your application.

Code or HTTP Excerpt

```
134 |     }
135 |
>>> 136 |     const filePath = join(UPLOADS_DIR, file.stored_name);
137 |     if (!existsSync(filePath)) {
138 |         return res.status(404).json({ error: 'File not found on disk' });
```

Business Impact

This vulnerability could be exploited to compromise application security and potentially access sensitive data.

Fix Recommendation

Rule-based recommendation: Review the flagged code and apply security best practices. Consult the OWASP guidelines for specific remediation steps relevant to this type of vulnerability.

LLM recommendation: Review the flagged code and apply security best practices. Consult the OWASP guidelines for specific remediation steps relevant to this type of vulnerability.

Appendix

Deduplication Statistics

Total raw findings: 6 | After deduplication: 3 | Reduction: 50%

Scan Metadata

Scan ID	e961216c-21ff-4d46-8ca5-4f548f6e2b7e
Target	C:\Users\Dhruv\Desktop\College Stuff\Projects\Messenger app
Timestamp	2026-07-23T13:49:51.542282+00:00
Duration	14.0s
Scanners Used	semgrep
Scanner Versions	semgrep=1.159.0

Raw Findings Data (JSON)

```
[
  {
    "id": "458f25ca-d763-415f-8495-80d46de02cab",
    "title": "Hardcoded Secret in C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\Messenger app\\backend\\nexus-data.json",
    "file": "C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\Messenger app\\backend\\nexus-data.json",
    "line_start": 7,
    "line_end": 7,
    "severity": "HIGH",
    "vuln_type": "HARDCODED_SECRET",
    "description": "bcrypt hash detected",
    "rule_id": "generic.secrets.security.detected-bcrypt-hash.detected-bcrypt-hash",
    "confidence": 0.9,
    "cwe_id": "CWE-798: Use of Hard-coded Credentials",
    "cvss_score": null,
    "code_snippet": "5 |         \\\"username\\\": \\\"alice\\\",\\n        6 |         \\\"email\\\": \\\"alice@test.com\\\",\\n",
    "sources": [
      "semgrep"
    ],
    "risk_score": 0.5488,
    "risk_category": "MEDIUM",
    "category_source": "score",
    "category_reason": "score-threshold",
    "llm_explanation": "Hardcoded secrets (passwords, API keys, tokens) in source code are a critical security vulnerability.",
    "fix_suggestion": "Move all secrets to environment variables. Use python-decouple or python-dotenv to load secrets.",
    "business_impact": "Exposed credentials could allow unauthorized access to external services, databases, or APIs.",
    "raw": {
      "check_id": "generic.secrets.security.detected-bcrypt-hash.detected-bcrypt-hash",
      "path": "C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\Messenger app\\backend\\nexus-data.json",
      "start": {
        "line": 7,
        "col": 25,
        "offset": 157
      },
      "end": {
        "line": 7,
        "col": 85,
        "offset": 217
      },
      "extra": {
```

```

    "message": "bcrypt hash detected",
    "metadata": {
      "cwe": [
        "CWE-798: Use of Hard-coded Credentials"
      ],
      "category": "security",
      "technology": [
        "secrets",
        "bcrypt"
      ],
      "confidence": "LOW",
      "owasp": [
        "A07:2021 - Identification and Authentication Failures",
        "A07:2025 - Authentication Failures"
      ],
      "references": [
        "https://owasp.org/Top10/A07_2021-Identification_and_Authentication_Failures"
      ],
      "cwe2022-top25": true,
      "cwe2021-top25": true,
      "subcategory": [
        "audit"
      ],
      "likelihood": "LOW",
      "impact": "HIGH",
      "license": "Semgrep Rules License v1.0. For more details, visit semgrep.dev/legal/rules-license",
      "vulnerability_class": [
        "Hard-coded Secrets"
      ],
      "source": "https://semgrep.dev/r/generic.secrets.security.detected-bcrypt-hash.detected-bcrypt-hash",
      "shortlink": "https://sg.run/3A8G"
    },
    "severity": "ERROR",
    "fingerprint": "requires login",
    "lines": "requires login",
    "validation_state": "NO_VALIDATOR",
    "engine_kind": "OSS"
  }
}
},
{
  "id": "098d2df9-43ca-4714-90ad-c4ac17ab4af2",
  "title": "Hardcoded Secret in C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\Messenger app\\frontend\\release\\win-unpack\\win-unpack.exe",
  "file": "C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\Messenger app\\frontend\\release\\win-unpack\\win-unpack.exe",
  "line_start": 7,
  "line_end": 7,
  "severity": "HIGH",
  "vuln_type": "HARDCODED_SECRET",
  "description": "bcrypt hash detected",
  "rule_id": "generic.secrets.security.detected-bcrypt-hash.detected-bcrypt-hash",
  "confidence": 0.9,
  "cwe_id": "CWE-798: Use of Hard-coded Credentials",
  "cvss_score": null,
  "code_snippet": "5 |         \"username\": \"alice\",\\n        6 |         \"email\": \"alice@test.com\"\\n    }\\n\\n    \"raw\": {\\n      \"check_id\": \"generic.secrets.security.detected-bcrypt-hash.detected-bcrypt-hash\",\\n      \"path\": \"C:\\\\Users\\\\Dhruv\\\\Desktop\\\\College Stuff\\\\Projects\\\\Messenger app\\\\frontend\\\\release\\\\win-unpack\\\\win-unpack.exe\",\\n      \"start\": {\\n        \"line\": 7,\\n        \"col\": 25,\\n        \"offset\": 157

```

```

    },
    "end": {
      "line": 7,
      "col": 85,
      "offset": 217
    },
    },
    "extra": {
      "message": "bcrypt hash detected",
      "metadata": {
        "cwe": [
          "CWE-798: Use of Hard-coded Credentials"
        ],
        "category": "security",
        "technology": [
          "secrets",
          "bcrypt"
        ],
        "confidence": "LOW",
        "owasp": [
          "A07:2021 - Identification and Authentication Failures",
          "A07:2025 - Authentication Failures"
        ],
        "references": [
          "https://owasp.org/Top10/A07_2021-Identification_and_Authentication_Failures"
        ],
        "cwe2022-top25": true,
        "cwe2021-top25": true,
        "subcategory": [
          "audit"
        ],
        "likelihood": "LOW",
        "impact": "HIGH",
        "license": "Semgrep Rules License v1.0. For more details, visit semgrep.dev/legal/rules-license",
        "vulnerability_class": [
          "Hard-coded Secrets"
        ],
        "source": "https://semgrep.dev/r/generic.secrets.security.detected-bcrypt-hash.detected-bcrypt-hash",
        "shortlink": "https://sg.run/3A8G"
      },
      "severity": "ERROR",
      "fingerprint": "requires login",
      "lines": "requires login",
      "validation_state": "NO_VALIDATOR",
      "engine_kind": "OSS"
    }
  }
},
{
  "id": "3d6e2e98-d061-46cd-8a15-89575844b116",
  "title": "Unknown in C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\Messenger app\\backend\\fileRoutes.js",
  "file": "C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\Messenger app\\backend\\fileRoutes.js",
  "line_start": 136,
  "line_end": 136,
  "severity": "MEDIUM",
  "vuln_type": "UNKNOWN",
  "description": "Possible writing outside of the destination, make sure that the target path is nested in",
  "rule_id": "javascript.express.security.audit.express-path-join-resolve-traversal.express-path-join-resolve-traversal",
  "confidence": 0.7,
  "cwe_id": "CWE-22: Improper Limitation of a Pathname to a Restricted Directory ('Path Traversal')",
  "cvss_score": null,
  "code_snippet": "      134 |       }\\n      135 | \\n>>> 136 |       const filePath = join(UPLOADS_DIR, file.s",
  "sources": [
    "semgrep"
  ],
  "risk_score": 0.3384,
  "risk_category": "LOW",
  "category_source": "score",
  "category_reason": "score-threshold",
  "llm_explanation": "A security vulnerability was detected in your code. This issue could potentially be",
  "fix_suggestion": "Review the flagged code and apply security best practices. Consult the OWASP guidelin",
  "business_impact": "This vulnerability could be exploited to compromise application security and potent"
}

```

```

"raw": {
  "check_id": "javascript.express.security.audit.express-path-join-resolve-traversal.express-path-join-traversal",
  "path": "C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\Messenger app\\backend\\fileRoutes.js",
  "start": {
    "line": 136,
    "col": 40,
    "offset": 4383
  },
  "end": {
    "line": 136,
    "col": 56,
    "offset": 4399
  },
  "extra": {
    "message": "Possible writing outside of the destination, make sure that the target path is nested in the destination",
    "metadata": {
      "owasp": [
        "A05:2017 - Broken Access Control",
        "A01:2021 - Broken Access Control",
        "A01:2025 - Broken Access Control"
      ],
      "cwe": [
        "CWE-22: Improper Limitation of a Pathname to a Restricted Directory ('Path Traversal')"
      ],
      "category": "security",
      "references": [
        "https://owasp.org/www-community/attacks/Path_Traversal"
      ],
      "technology": [
        "express",
        "node.js"
      ],
      "cwe2022-top25": true,
      "cwe2021-top25": true,
      "subcategory": [
        "vuln"
      ],
      "likelihood": "HIGH",
      "impact": "MEDIUM",
      "confidence": "MEDIUM",
      "license": "Semgrep Rules License v1.0. For more details, visit semgrep.dev/legal/rules-license",
      "vulnerability_class": [
        "Path Traversal"
      ],
      "source": "https://semgrep.dev/r/javascript.express.security.audit.express-path-join-resolve-traversal",
      "shortlink": "https://sg.run/weRn"
    },
    "severity": "WARNING",
    "fingerprint": "requires login",
    "lines": "requires login",
    "validation_state": "NO_VALIDATOR",
    "engine_kind": "OSS"
  }
}
}
]

```